International Institute of Information Technology, Hyderabad

RAG over the EU AI Act

Baseline Retrieval vs. Cross-Encoder Re-Ranking

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RAG Assignment

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Code: `https://github.com/vishakkashyapk30/lma-rag-assignment`

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Setup

Document	The EU AI Act (Regulation 2024/1689), the official 144-page PDF. It has bans with narrow exceptions, duties that change with the role (provider, deployer, importer), and amounts that are easy to mix up. It also has 180 recitals. These explain the law in plain words but are not binding, so they make natural distractors. Code: `https://github.com/vishakkashyapk30/lma-rag-assignment`
Chunking	I removed headers, footers and footnotes. Each recital is one chunk. Article paragraphs and annex points are packed up to 230 tokens (the bi-encoder reads 256). Each chunk starts with a header like `[Article 99 - Penalties, para 5-6]`. That gives 747 chunks: 289 recital, 387 article and 71 annex.
Pipelines	*Baseline:* the bi-encoder takes the top 10 chunks by cosine, the top 3 go to the LLM, giving Answer 1. *Re-ranking:* the cross-encoder scores the same 10 (question, chunk) pairs and re-sorts them, and the new top 3 go to the LLM with the identical prompt, giving Answer 2. *Control:* the LLM answers with no chunks. *Final pipeline:* an improved version (Section 5).
Models	Bi-encoder `all-MiniLM-L6-v2` with an exact FAISS inner-product index. Cross-encoder `ms-marco-MiniLM-L-6-v2`. LLM `qwen2.5:7b` through Ollama at temperature 0, on an RTX 4060 laptop GPU.
Questions	49 questions, written the way a user would ask them. They cover negation, exception, corner case, role limit, keyword trap, scope and plain questions (all listed on the last page). Each has a gold clause, matched by article, paragraph and a key phrase, because Article 5(1) alone has six chunks. I used Q01 to Q24 to choose settings and Q25 to Q49 only to report.
Grading	Retrieval: hit@3 (a right clause in the top 3), all@3 (every needed clause in the top 3) and MRR@10. Answers: gpt-4.1 graded each one correct (1), partial (0.5) or wrong (0) against the law text. A second model, gpt-4.1-mini, graded blind. The two agree exactly on 84% of answers and within one step on 100%. I also give a string check of key facts.

Table 1: Setup of the experiment.

1 Architecture Comparison

Both models use the same 6-layer MiniLM transformer. The only difference is how the question and the chunk go in (Figure 1). Figure 2 shows where each model sits in the two pipelines. A **bi-encoder** encodes them separately and mean-pools each into one vector, then compares the two with cosine similarity: $s(q, d) = \cos(E(q), E(d))$. The chunk vectors are made once and stored, so search is fast. But the question and the chunk never meet, and a word like "not" or "importer" gets mixed in with about 200 others. A **cross-encoder** joins them into one sequence. Self-attention lets every question token look at every chunk token, and a linear layer on the `[CLS]` vector gives one score: $s(q, d) = \mathbf{w}^\top \mathbf{h}_{[\mathrm{CLS}]}(\mathrm{MiniLM}([\mathrm{CLS}]\, q\, [\mathrm{SEP}]\, d\, [\mathrm{SEP}]))$. It can see conditions and negation. The catch is that nothing can be precomputed, because every pair needs its own pass (Table 2).

A real example Q16 asks what an importer must check. The bi-encoder gave paragraphs 2 to 4 of Article 23 a cosine of 0.746 and put them first. Paragraph 1, which lists the checks, got 0.680 and came fourth. The cross-encoder gave paragraph 1 a score of 8.17 and paragraphs 2 to 4 a score of 6.84, so paragraph 1 moved to first place. All of them are about importers. I think the cross-encoder did better because it reads the question and the paragraph together. It can see what the paragraph tells the importer to do, not only its topic.

2 Qualitative & Rank Shift Analysis

Rank changes Re-ranking helped retrieval only a little. MRR@10 went from 0.474 to 0.502 and hit@3 from 0.633 to 0.653 (Table 3). The right clause moved up for 15 questions and down for 8. Four clauses moved from rank 4 or 5 into the top 3 (Q16 $\#4 \to \#1$, Q02 $\#4 \to \#2$, Q27 $\#5 \to \#2$, Q47 $\#5 \to \#2$), and three moved out of it (Q20 $\#1 \to \#5$, Q46 $\#2 \to \#9$, Q44 $\#2 \to \#4$). Figure 3 shows why. Over the 490 candidates the two models agree on the broad order (Spearman 0.72), but the cross-encoder pushed about a third of the bi-encoder's top 3 chunks out of the top 3. Figure 4 shows the whole top 10 for six questions, and Table 4 groups the questions by the failure types from the assignment.

Answers, rescued cases In the four rescued questions the right clause sat at rank 4 or 5, so the baseline LLM never saw it. For Q16 the baseline answer got the rule wrong. It said the importer must make sure the system "is not in conformity". For Q02 and Q27 it had only a recital and left out an essential clause ("untargeted scraping", and the exception for lawful dataset labelling). After re-ranking, Q02 was correct and Q16 partial. Q27 stayed partial, though the second judge marked it correct. Q47 was already right, because a recital says almost the same thing. The judge labelled none of the baseline answers as made up. Reading them myself, I found one invented sentence in the re-ranked answer to Q23, which said the 10 year rule is "similar for providers and importers". The judge gave that answer full marks because the number is right.

Keyword trap and distractor Q47 is the clearest case. Recital 58 uses all 8 content words of the question, and the real rule in Annex III uses 7. The bi-encoder put the recital first and the annex fifth. The cross-encoder moved the annex to second. Q20 is the opposite, a different-wording trap. The law says "interact directly with natural persons" and the question says "chatting with a bot". They share 1 of 6 words, and re-ranking made it worse. I do not count Q02 as a keyword trap, because its right clause shares as many words with the question as the wrong one. Recitals trick both models. The top chunk was a recital for 25 questions before re-ranking and for 27 after. This is because recitals explain in plain words but often leave out the exact condition or number.

Where re-ranking did not help *Regressions.* The cross-encoder gets it wrong too. For Q20, Q46 and Q44 it pushed the right clause out of the top 3. *Lost chunks.* In 11 questions the right clause was not in the top 10, so the cross-encoder never saw it. Q11 is the worst. The top 10 had the general fine, "whichever is higher", but not the start-up cap, "whichever is lower", so the re-ranked answer gave the wrong rule. *Reading failures.* The baseline had the right clause in its top 3 for 31 questions, and the answer was still not fully correct for 5. For re-ranking it was 32 and 8. Q33 is one example. The law says an AI-written article needs no label when an editor is responsible, and the answer said it does. Better retrieval cannot fix that.

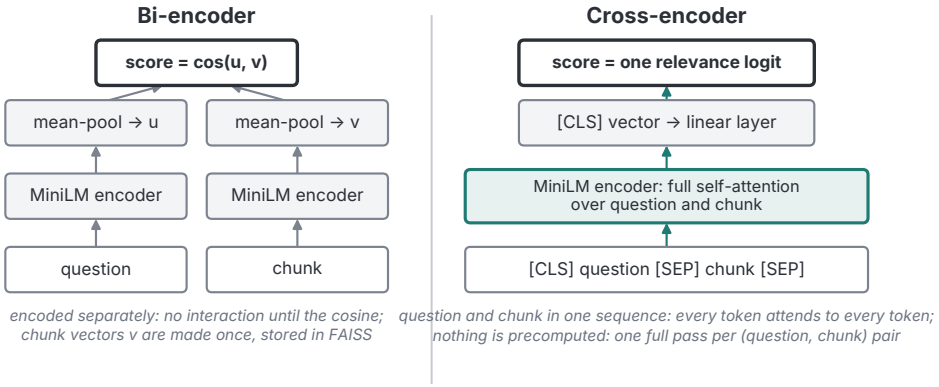


Figure 1: The same MiniLM transformer used in two ways. The cross-encoder path is in green.

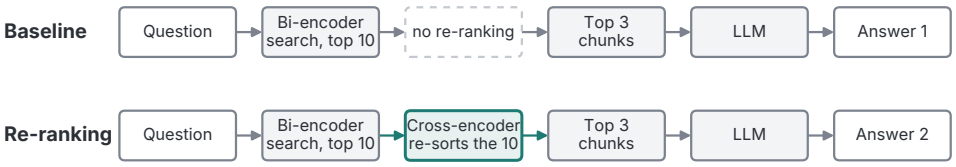


Figure 2: The two pipelines. Both start with the same bi-encoder search. The re-ranking pipeline adds the cross-encoder, which reads the question with each of the 10 chunks and keeps the best 3. The baseline just keeps the first 3 by cosine.

	Bi-encoder	Cross-encoder
Input and interaction	question and chunk encoded separately, no interaction until the final cosine	one joined sequence, every token attends to every token
Output and precompute	two 384-dimensional vectors; all chunk vectors made once	one relevance score; nothing can be precomputed
Cost per question	one encoder pass plus dot products	one encoder pass for every chunk
Trained to	put sentences with similar meaning close together	judge whether a passage answers a question (MS MARCO labels)
Good at	finding the right topic fast	fine detail: conditions, exceptions, negation

Table 2: What the architecture difference means in practice.

	First stage	Re-ranker	hit@3	all@3	MRR@10	MRR test	Recall	Key facts	Grade	Grade 2	ms
No retrieval	–	–	–	–	–	–	–	–	0.265	0.337	–
Baseline	bi-encoder	none	0.633	0.612	0.474	0.471	0.776	0.673	**0.735**	0.684	4.5
Re-ranking	bi-encoder	MiniLM-L6	0.653	0.633	0.502	0.498	0.776	0.769	0.724	0.694	21.5
Final pipeline	bi-enc. + BM25	MiniLM-L12	**0.837**	**0.796**	**0.818**	**0.825**	**0.980**	**0.810**	0.694	**0.735**	87.4

Table 3: The three setups and the no-retrieval control on all 49 questions. MRR test = MRR@10 on Q25 to Q49. Recall = share of questions whose right clause was among the candidates (the top 10, or 30 for the final pipeline). Key facts = share of key facts found in the answer. Grade and Grade 2 = mean grade from gpt-4.1 and from gpt-4.1-mini. ms = median retrieval time. The best value in each column is in bold.

Is it only the LLM's knowledge? No. Without chunks the mean grade was 0.265, against 0.735 with the baseline (Figure 5). Only 1 of 49 answers was fully correct, and none said that it did not know. The model just made up details. For Q11 it gave €30 million or 6%, where the law says €35 million or 7%. Re-ranking did not clearly raise the grades. gpt-4.1 gives 0.735, 0.724 and 0.694 to the baseline, re-ranking and final pipeline. gpt-4.1-mini gives 0.684, 0.694 and 0.735. The answer got better for only 2 of the 15 questions whose right clause moved up. Better retrieval did not clearly give better answers.

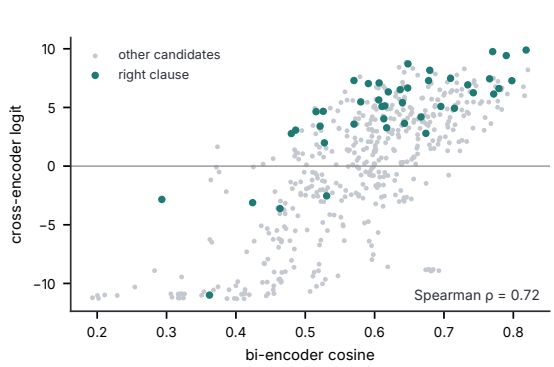
3 Latency Profiling

I timed each stage per question after 5 warm-up runs. I ran every question 3 times, with `cuda.synchronize()` around each call (Table 6).

Stage (ms)	median	p95
Embed the question	4.35	4.65
FAISS search	0.11	0.14
Cross-encoder, 10 pairs, GPU	17.01	17.73
Cross-encoder, 10 pairs, CPU	120.91	140.18
Baseline retrieval	**4.47**	**4.75**
Re-ranking retrieval, GPU	**21.48**	**22.32**
Re-ranking retrieval, CPU	**125.38**	**144.77**
LLM answer (both)	5,416	9,460
Search + re-rank, one at a time	19.1	–
Search + re-rank, 49 questions batched	11.2	–

Table 6: Time per question for each stage.

Re-ranking adds about 17 ms on the GPU, so retrieval goes from 4.5 ms to 21.5 ms, about 5 times slower. On the CPU it adds about 121 ms, 28 times slower. Almost all of that is the cross-encoder's 10 passes. The LLM takes about 5 seconds, so the extra time is under 1% of the wait. Scoring the pairs of



(a) Bi-encoder score vs. cross-encoder score.



(b) Bi-encoder rank (rows) vs. rank after re-ranking.

Figure 3: The two models on the same 490 candidates (49 questions, 10 each). Gold chunks are in teal. In (b) the outlined block is the top 3 in both rankings.

Failure type	Questions	hit@3, before → after	Fixed	Worse	Not in top 10	Example
Keyword trap (as labelled)	4	1 → 2	1	0	2	Q02 #4 → #2. Q18 was already first. Q19 and Q49 (dates) were never retrieved
Different wording (synonym)	1	1 → 0	0	1	0	Q20: "chatting with a bot" vs. "interact directly with natural persons", #1 → #5
Exceptions and corner cases	18	9 → 9	1	1	5	Q47: fraud exception, and the clearest distractor, #5 → #2. Q11: the start-up cap (Art 99(6)) never reached the top 10
Negation and constraints	7	4 → 6	2	0	1	Q27 #5 → #2 and Q02 #4 → #2
Role limits	9	8 → 8	1	1	0	Q16 importer #4 → #1. Q44 got worse, #2 → #4
Buried at rank 4 to 10	4	0 → 4	4	0	0	Q16, Q02, Q27, Q47 all moved into the top 3

Table 4: Results for each failure type. Fixed = right clause outside the top 3 before and inside after. Worse = the other way round. Not in top 10 = re-ranking could not help. A question can have more than one type (Q47 is both a distractor and an exception), so the rows do not add up.

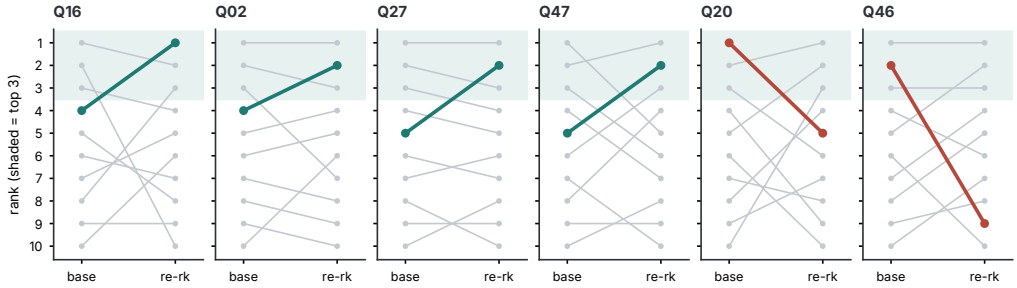


Figure 4: Rank of every top 10 chunk before (base) and after re-ranking (re-rk). The right clause is in colour: green moved up, red moved down. Q16, Q02, Q27 and Q47 were rescued. Q20 and Q46 got worse.

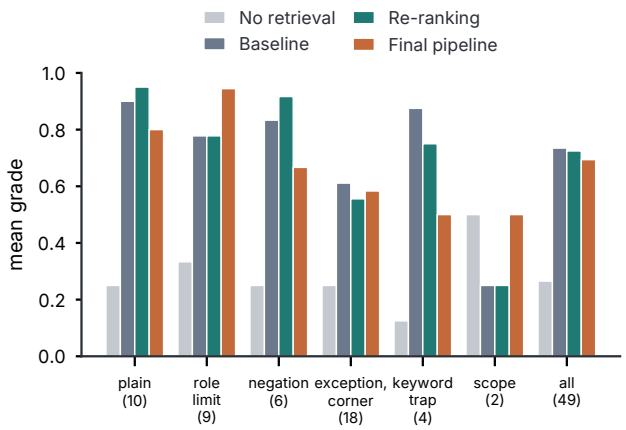


Figure 5: Mean grade of the answers by kind of question (gpt-4.1). The number in brackets is the number of questions.

all 49 questions in one batch costs about 11 ms per question instead of 19 ms.

4 Scaling Question

A cross-encoder needs the question, so nothing can be precomputed. For every new question it must run one pass for every chunk. I ran it on all 747 chunks for one question, and it took 870 ms, or 1.16 ms per chunk, about 190 times slower than the 4.6 ms of the whole bi-encoder search. With 1 million chunks it would need about 20 minutes per question (Figure 6). The bi-encoder embeds the chunks once, offline, and an approximate index keeps the search in milliseconds. The Sentence-BERT paper has the classic example: the best pair among 10,000 sentences takes 65 hours with a BERT cross-encoder and 5 seconds with SBERT.

Question	Type	Right clause	Baseline answer (grade, label)	Re-ranked answer (grade, label)
Q16 importer checks	role limit	#4 → #1	0, misstates the rule	0.5, omits an essential clause
Q02 scraping photos	negation	#4 → #2	0.5, omits an essential clause	1, correct
Q27 sort by race	negation	#5 → #2	0.5, omits an essential clause	0.5, omits an essential clause
Q47 fraud exception	exception	#5 → #2	1, correct	1, correct
Q20 chatting with a bot	wording	#1 → #5	1, correct	1, correct
Q46 missing child	exception	#2 → #9	1, correct	0.5, omits an essential clause
Q44 lie detection	role limit	#2 → #4	1, correct	0.5, omits an essential clause
Q23 keeping documents	role limit	not in top 10	0, misstates the rule	1, correct
Q11 start-up fine	corner case	cap clause not in top 10	0, gave up	0, misstates the rule

Table 5: Baseline vs. re-ranked answer for the failure cases, with the label from gpt-4.1. The first four are the rescued questions. For Q11 the rank is for the start-up cap clause. For Q23 neither pipeline retrieved the right clause, and re-ranking promoted the importer's 10 year rule.

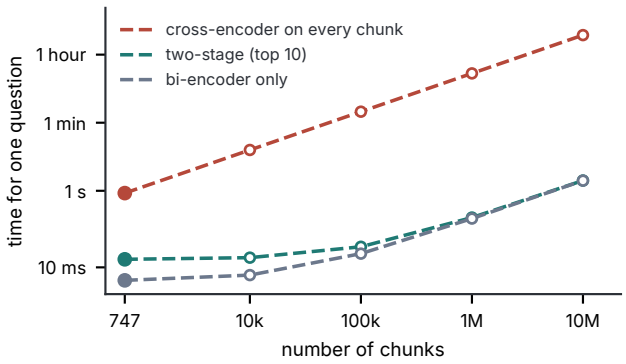


Figure 6: Time for one question as the collection grows. The first point is measured and the rest are scaled up linearly.

This is why retrieval is done in two stages. The cheap model narrows the collection to a few candidates, and the expensive one orders only those. The weakness is that re-ranking can only fix what the first stage found. When I let the cross-encoder score all chunks, it moved Q23 from rank 41 to 1, Q06 from 16 to 1 and Q26 from 14 to 1.

5 Ablation Study and Improved Retrieval Pipeline

Two limits showed up. The cross-encoder cannot see what the first stage missed, and recitals trick both models. So I changed one design choice at a time and measured retrieval only (Table 7).

Change	MRR	Recall	Dev	Test
Reference (re-ranking)	0.502	0.776	0.506	0.498
Chunks of at most 120 tokens	0.558	0.796	0.577	0.539
Chunks of at most 400 tokens	0.444	0.653	0.503	0.387
No chunk header	0.496	0.735	0.505	0.487
Recitals removed	0.736	0.837	0.748	0.725
Re-rank the top 50	0.552	0.898	0.581	0.525
Add BM25 to the first stage	0.563	0.857	0.585	0.543
Bigger bi-encoder (bge-base)	0.560	0.796	0.569	0.551
Bigger re-ranker (MiniLM-L12)	0.537	0.776	0.545	0.530
All best changes (final pipeline)	**0.818**	**0.980**	**0.812**	**0.825**

Table 7: One change at a time. MRR = MRR@10 after re-ranking. Recall = share of questions whose right clause reached the cross-encoder. Dev and Test = MRR on Q01 to Q24 and Q25 to Q49.

Chunks of at most 120 tokens helped (0.558 against 0.502). Chunks of 400 tokens hurt (0.444), because the bi-encoder cut 216 of 467 chunks at 256 tokens. Removing the *header* lowered first-stage hit@3 from 0.633 to 0.469. *Removing the recitals* gave the biggest single gain (0.736). Recitals take places in the top 10 that the article should have. My labels point to the law and never to a recital, so some of this gain was expected. A *longer candidate list* (top 50) raised Recall from 0.776 to 0.898. *BM25* merged with the bi-encoder by reciprocal rank fusion (a chunk gets $1/(60 + \text{rank})$ from each list) raised first-stage hit@3 from 0.633 to 0.694, because it catches exact words like "start-ups". A bigger bi-encoder or re-ranker alone helped little (0.560, 0.537).

The **final pipeline** indexes only articles and annexes, merges the bi-encoder and BM25 rankings into 30 candidates, re-ranks them with MiniLM-L12, and sends the best 3 to the same prompt. Recall rose to 0.980 and MRR@10 to 0.818. On the held-out questions it is 0.825, against 0.812 on the dev questions, so it is not fitted to them. A larger re-ranker (bge-reranker-v2-m3) reached 0.828 but did not fit on the GPU next to the LLM. The final pipeline still fails on Q11, because the cross-encoder put Article 99(6) at rank 8. The key facts rose from 0.673 to 0.769 to 0.810, but the graded answers did not follow (0.735, 0.724, 0.694). I did not test why. It could be reading failures, the plain text of the recitals that it no longer uses, or judge noise.

6 Summary

Key points

- Re-ranking fixed four buried clauses and broke three cases. It cannot recover a clause that the first stage missed (11 of 49 questions).
- Recitals and first-stage recall mattered more than the choice of re-ranker. Removing recitals and adding BM25 moved MRR@10 from 0.50 to 0.82.
- Retrieval matters, since the mean grade was 0.27 without it and 0.73 with it. But re-ranking and the final pipeline did not clearly raise the graded answers.

Limits The two LLM judges disagree on the order of the three setups, so I do not claim that one is better for answers. The key fact check is rough (Q02 scores zero everywhere, although its answers are right). I also used only 49 questions and one LLM.

References

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List of questions

The rank of the best right chunk for each question. "–" means it was not in the top 10. Base is the baseline (bi-encoder only), Re-rk is re-ranking the top 10, and Final is the final pipeline.

ID	Question	Gold	Base	Re-rk	Final
Q01	Is it allowed to use AI to detect the emotions of employees at their workplace?	Art 5(1)	3	3	1
Q02	Is it legal to build a face recognition database by scraping photos from the internet?	Art 5(1)	4	2	1
Q03	Can an AI system be used to predict whether a person will commit a crime?	Art 5(1)	1	2	1
Q04	When are police allowed to use real-time remote biometric identification in public spaces?	Art 5(1)	–	–	8
Q05	Can real-time biometric identification be used before a judge or authority has approved it?	Art 5(3)	4	4	2
Q06	Is every AI system listed in Annex III automatically considered high-risk?	Art 6(3)	–	–	1
Q07	Do providers of open-source general-purpose AI models have to prepare technical documentation?	Art 53(2)	5	5	9
Q08	Do deepfakes have to be labelled even when they are part of a satirical or artistic work?	Art 50(4)	3	2	1
Q09	Does the AI Act apply to AI systems used only for military purposes?	Art 2(3)	3	3	1
Q10	Does a person using an AI system for a personal hobby have obligations under the AI Act?	Art 2(10)	–	–	4
Q11	What is the maximum fine for a start-up that puts a prohibited AI system on the market?	Art 99(3) + Art 99(6)	1	1	1
Q12	What is the fine for giving misleading information to the authorities?	Art 99(5)	1	1	1
Q13	What fines can be imposed on providers of general-purpose AI models?	Art 101(1)	1	1	1
Q14	What are the obligations of a deployer, as opposed to a provider, of a high-risk AI system?	Art 26	2	1	1
Q15	When is a distributor or deployer treated as the provider of a high-risk AI system?	Art 25(1)	1	1	1
Q16	What must an importer check before placing a high-risk AI system on the market?	Art 23(1)	4	1	1
Q17	Which deployers must carry out a fundamental rights impact assessment?	Art 27(1)	3	3	2
Q18	How much training compute makes a general-purpose AI model count as having systemic risk?	Art 51(2)	1	1	1
Q19	From what date do the bans on unacceptable AI practices start to apply?	Art 113	–	–	–
Q20	Do people have to be told that they are chatting with a bot?	Art 50(1)	1	5	1
Q21	Is an AI system used for credit scoring considered high-risk?	Annex III pt 5	3	2	1
Q22	Does a person affected by a decision made with a high-risk AI system have a right to an explanation?	Art 86(1)	1	2	1
Q23	For how long must providers keep the technical documentation of a high-risk AI system?	Art 18(1)	–	–	1
Q24	What is the definition of an AI system?	Art 3(1)	3	1	1
Q25	Are SMEs and start-ups given lower fines under the AI Act?	Art 99(6)	3	1	1
Q26	Is it prohibited to use AI that exploits the vulnerabilities of children or elderly people?	Art 5(1)	–	–	1
Q27	Can a company use biometric data to sort people by their race or sexual orientation?	Art 5(1)	5	2	1
Q28	Is social scoring of citizens banned?	Art 5(1)	3	2	1
Q29	How quickly must a provider report a serious incident that caused a person's death?	Art 73(4)	1	1	1
Q30	What is the deadline for reporting a serious incident?	Art 73(2)	3	2	1
Q31	Does an employer have to inform workers before using a high-risk AI system in the workplace?	Art 26(7)	3	2	1
Q32	Can a high-risk AI system be used without completing the conformity assessment?	Art 46(1)	–	–	6
Q33	Do AI-generated news articles need to be disclosed if a human editor reviewed them?	Art 50(4)	1	2	1
Q34	Must people be informed when an emotion recognition system is used on them?	Art 50(3)	3	1	1
Q35	Is an AI tool that screens job applications and filters CVs high-risk?	Annex III pt 4	1	1	1
Q36	Does the AI Act apply to AI being developed and tested before it is released?	Art 2(8)	–	–	2
Q37	What happens to high-risk AI systems already on the market before the rules apply?	Art 111(2)	1	1	1
Q38	Does the deepfake disclosure obligation apply when the content is used by law enforcement to investigate crimes?	Art 50(4)	2	1	1
Q39	Are there exceptions to the ban on emotion recognition at work?	Art 5(1)	5	4	1
Q40	Can a provider avoid the high-risk obligations by documenting that its Annex III system is not risky?	Art 6(3) + Art 6(4)	–	–	1
Q41	What fine does a deployer face for not complying with its obligations?	Art 99(4)	1	1	1
Q42	Is AI used to assess students' exams or admission considered high-risk?	Annex III pt 3	1	2	1
Q43	Can AI be used to decide on asylum applications?	Annex III pt 7	1	3	1
Q44	Which uses of AI by law enforcement for lie detection are high-risk?	Annex III pt 6	2	4	2
Q45	Is facial recognition allowed for unlocking a phone or verifying identity?	Annex III pt 1	–	–	4
Q46	Can police use biometric identification to find a missing child?	Art 5(1)	2	9	5
Q47	Are AI systems used for detecting financial fraud classified as high-risk?	Annex III pt 5	5	2	1
Q48	How long do providers have to keep logs automatically generated by a high-risk AI system?	Art 19(1)	1	1	1
Q49	When will most of the AI Act's rules start to apply?	Art 113	–	–	–